

Math is Fun!

A tiny adventure in OllieCode - For kids about 7 years old

What you will make

You will teach Ollie the robot to play a math game:

- Ollie picks two secret numbers, a and b.
- Ollie asks you: *What is a + b?*
- You type your answer.
- Ollie says "You're right!" or "Try again!"

That is your very own Math is Fun app.

Before you start

- Ask a grown-up to open OllieCode on the computer.
 - Find the blocks on the left and the stage (the picture where Ollie lives) on the right.
 - Take your time. It is okay to make mistakes - we can fix them.
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Step 1 - Wake up the program

- 1. Look in the toolbox for Events.
- 2. Drag out the block that says When Run clicked.
- 3. This block means: `"Start here when I press the Run button."`

Everything else will hang under this block like a necklace.

Step 2 - Do the game more than once

- 1. Open Control.
 - 2. Find repeat ... times and set it to 10 if you want ten rounds.
 - 3. Put repeat right under When Run clicked.
 - 4. All the next steps go inside the repeat - in the mouth of the repeat block.
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Step 3 - Pick two surprise numbers

- 1. Open Variables and Math.
- 2. Make two variables named a and b (the grown-up can help with Create variable).
- 3. Under repeat, add:
- set a to ... a random integer from 0 to 10 (or numbers you like).

- set b to ... another random integer.

Now a and b are new numbers every round. That keeps the game fun.

Step 4 - Ask the question and remember the answer

- 1. Open Variables and Text.
- 2. Make a variable named answer (or guess - pick one name and stick with it).
- 3. Add set answer to ... and plug in ask ... and wait for number from Text.
- 4. In the ask block, put your question. You can type:
 - a simple line like What is a + b?, or
 - use create text with to build a sentence with a, +, and b so the numbers show up.

When you run the program, you type a number in the box. That number is saved in answer.

Step 5 - The magic check (read this twice!)

This is the part that tricks a lot of people. Here is the secret:

You must check whether your typed number is the same as a + b.

Do not compare two names that mean the same thing in your head but are not set in the computer.

Do this instead:

- 1. Open Logic and use if / else (or if with two parts).
- 2. In the if, use the equals test.
- 3. On the left, put the answer variable (the round block with answer).
- 4. On the right, put Math -> add -> plug in a and b.

So the question in the computer is really:

Is answer the same as a + b?

If you compare guess to answer but you never set guess to anything, the game will feel broken. The computer only knows what you set with your blocks.

Step 6 - What Ollie says

- 1. Open Looks.
- 2. Under if, put say ... for ... seconds - for example You're right!
- 3. Under else, put another say - for example Try again!

You can pick your own words. Keep them short so you can read them on the stage.

Step 7 - Press Run and play

- 1. Tap Run.
- 2. Read Ollie's question.

- 3. Type your number and tap OK.
- 4. Watch the speech bubble.

Play a few times. Change a and b ranges when you are ready for harder math.

Extra fun (optional)

- Add play sound under When Run clicked for happy music.
 - Change the scene behind Ollie so your game has a cool background.
 - Use Save and name your project Math is Fun so you can open it again.
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If something feels wrong

What happens	What to try
It always says "Try again"	Check Step 5: left side = answer, right side = $a + b$.
It always says "You're right"	You might be comparing the wrong things. Ask a grown-up to read Step 5 with you.
The ask box never shows	Make sure set answer to and ask are inside repeat, under When Run clicked.

You did it. You built a real quiz. Ollie is proud of you.

OllieCode - Math is Fun lesson

OllieCode - Math is Fun lesson (age 7)